

Code-Layout, Readability and Reusability

Date	25 May 2026
Team ID	XXXXXX
Project Name	Smart Lender – AI Loan Approval Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	yes	Proper indentation maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	yes	Files are organized into templates, static, model, and backend folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	Variables use clear and descriptive names.
4	Function / Method Names	Functions are descriptively named	yes	Functions are named based on their purpose.
5	Code Comments	Inline and block comments explain logic	yes	Important functions include comments for better understanding.
6	Modular Design	Code is split into reusable functions/modules	yes	Code is divided into reusable modules and functions.
7	No Redundant Code	Duplicate or unused code is removed	yes	Duplicate and unused code has been removed.
8	Error Handling	Exceptions and errors are handled gracefully	yes	Input validation and exception handling are implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, Flask	Backend	High

2	Loan Prediction Model	Scikit-learn, XGBoost	Prediction Module	High
3	Data Preprocessing	Pandas, NumPy	Model Training & Prediction	High
4	HTML Templates	HTML, Bootstrap	User Interface	High
5	CSS & JavaScript	CSS, JavaScript	Frontend Pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure.
Readability	5	Clean, simple, and easy-to-understand code.
Reusability	5	Reusable functions and modules improve maintainability.
Documentation / Comments	5	Key sections are documented with comments.
Overall Score	5	The Smart Lender project follows good coding practices and is modular, readable, and maintainable.